

A Systematic Framework for Backtesting and Live Paper Trading of Quantitative Cryptocurrency Strategies

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Abstract

Cryptocurrency markets trade 24/7 with high volatility, fat-tailed returns and material transaction costs, making the design and *honest* evaluation of systematic strategies difficult. This paper presents an end-to-end Python framework that unifies data acquisition, indicator engineering, strategy implementation, vectorised and event-driven backtesting, risk/execution modelling, performance analytics and a real-time paper/live engine. The strategy set spans a standard *intermediate & advanced* crypto strategy strategy set: Ichimoku Cloud, calendar anomalies, Aroon/RSI divergence, time-series and cross-sectional momentum, cointegrated pairs trading, K-Means market clustering and Hurst-exponent regime filtering.

The framework is validated with 53 unit tests that enforce causality (no look-ahead) and demonstrated on real Binance data (BTC/USDT 1h, 4,319 bars, 180 days). With zero-cost assumptions removed and 0.1% per-side fees plus slippage applied, every single-symbol strategy underperforms buy-and-hold in that sample, which we report transparently. We then run a 384-configuration selection sweep (timeframe x Ichimoku preset x cross window x exit rule x stop width) over three years of BTC/USDT and choose on out-of-sample behaviour rather than on the best in-sample cell: removing the fixed take-profit and moving from 1h to 4h turns the Ichimoku system from -33.9% into +19.3% over the same three years, with maximum drawdown cut from -37.2% to -13.0% (buy-and-hold: +197.3%, -53.5%). That gain is concentrated in very few trades - the single best trade equals the entire three-year profit - so we report it as a cost-and-robustness finding rather than a proven edge. The K-Means-clustered, long-only momentum portfolio also underperformed a BTC buy-and-hold (8,083.26 vs 9,574.27, -15.6%) while still beating its two weakest members (ADA, XRP). A paper/live engine ran overnight on BTC/USDT 1m and traded through an order-book-walking simulator with ATR stops; the full journal (paper + replay + live) records 57 round-trip trades and 116 fills. We argue the framework's contribution is validated *plumbing* — realistic costs, risk and causality — rather than a proven profitable signal, and outline the path to forward-testing ML-driven signals.

Keywords: algorithmic trading, cryptocurrency, Ichimoku, pairs trading, K-Means, Hurst exponent, event-driven backtesting, transaction costs.

1. Introduction

Retail and even institutional access to crypto exchange data is now trivial; what remains hard is turning that data into a *defensible* trading system. Three problems dominate:

1. **Look-ahead bias** — naive strategies inadvertently use future information, inflating backtest performance.
2. **Ignored costs** — ignoring fees, spread, slippage and market impact turns a "profitable" backtest into a losing account.
3. **Unrealistic fill modelling** — assuming fills at the signal bar's close rather than the ground truth of next-bar execution.

This project builds a framework in which these three problems are first-class citizens, and uses it to implement and evaluate the full set of systematic strategies. The contributions are:

- A **two-engine backtesting design**: a vectorised research engine (full-notional, proportional costs) for fast parameter sweeps, and an **event-driven engine** that simulates the live engine bar-by-bar (next-open fills, position sizing, ATR/Kumo stops, circuit breakers).
- A **library of gate-keeping modules** for fraud-proof evaluation: causal indicator implementations, a cost model (fees, spread, slippage, square-root impact), drawdown depth/duration statistics and multi-metric analytics (Sharpe, Sortino,

Calmar, PSR, VaR/CVaR, alpha/beta).

- A **real-time asyncio engine** that runs the same strategy objects on live WebSocket (ccxt.pro) or replayed data with paper (order-book walking) and CCXT-based testnet/live execution, all journaled to SQLite.
 - An **honest experimental protocol**: results are reported with costs included, negative sample results are published rather than hidden, and every strategy ships with causality tests.
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2. Related Work

2.1 Ichimoku trading systems

Ichimoku Kinko Hyo (☐☐☐) is a five-line Japanese indicator (Tenkan, Kijun, Senkou A, Senkou B, Chikou) widely used for trend and momentum. Quantitative treatments focus on Tenkan/Kijun crosses and Kumo (cloud) breakouts with an intrinsic trailing stop at the cloud edge.

2.2 Pairs trading & cointegration

Vidyamurthy (2004) and Gatev et al. (2006) formalise statistical arbitrage as trading a mean-reverting spread. The celebrated "classic" approach constructs a spread from an OLS hedge ratio and trades z-score thresholds of the rolling spread. We implement exactly this, using the ADF test (statsmodels) on the log-price spread to confirm stationarity before trading.

2.3 Market clustering & regime detection

K-Means clustering on market features is used both naively (classical state detection on volatility) and as an asset-classifier (clustering assets by volatility/momentum/liquidity then allocating into the best cluster). Hurst exponent (rescaled-range) classifies a series as trending ($H > 0.5$), mean-reverting ($H < 0.5$) or random-walk ($H = 0.5$); we combine it with RSI as a regime gate.

2.4 Momentum & portfolio construction

Cross-sectional momentum ranks assets by past returns and holds the top decile (Jegadeesh & Titman, 1993); equal-weight top-N portfolios avoid the estimation error of mean-variance weights. We implement a two-component ranking `rank(ret2)+rank(retL)- 2*rank(vol)` to penalise high-volatility "momentum trash".

3. Methodology

3.1 Data layer

- **Binance (CCXT)**: paginated OHLCV download with retry/back-off, gap detection, and an idempotent SQLite cache supporting incremental refresh.
- **Synthetic**: a regime-switching geometric-Brownian-motion generator (Markov chain over hidden volatility regimes) for offline research and unit tests.
- **CSV/Parquet**: round-trip persistence for reproducibility.

For the results in §5 we use real Binance BTC/USDT 1-hour OHLCV: **4,319 bars, 180 days, 2026-03-15 → 2026-09-11**, with a period return of **+8.95%** (buy-and-hold from the first to the last bar).

3.2 Indicators (`indicators.py`)

All indicators are vectorised and **strictly trailing** (past bars only):

- Ichimoku: five lines + Kumo (cloud) + projection; Wilder-smoothed ATR.
- RSI (Wilder) plus a **causal divergence detector** (compares rolling extrema of price vs RSI within a lookback window).
- Aroon (Chande), Bollinger Bands, rescaled-range **Hurst exponent** (R/S regression over lags) and the **ADF** stationarity test (statsmodels) with a version-safe API.

An adaptive Ichimoku variant labels each bar's volatility regime (Low/Normal/High) either by quantile thresholds or by **causal K-Means** on trailing NATR and switches between three Ichimoku parameter presets.

3.3 Strategies

#	Strategy	Module	Course
S1	Ichimoku 10/30/60/30 (TK cross + Kumo filter, long-only)	<code>strategy.py</code>	Intermediate
S2	Long-only time-series momentum (20-bar return, +2% entry / 0% exit, SMA filter)	<code>momentum.py</code>	Intermediate
S3	Calendar anomalies (hour & day-of-week seasonal profiles, expanding-means)	<code>intermediate_strategies.py</code>	Intermediate
S4	Aroon/RSI divergence (Aroon trend gate + RSI divergence entry)	<code>intermediate_strategies.py</code>	Intermediate
S5	Cointegrated pairs: BTC/USDT × ETH/USDT, rolling OLS ratio → z-score entries	<code>pairs.py</code>	Advanced
S6	Hurst regime (H>0.5 trend: RSI cross-50; H<0.5 reversion: RSI extremes)	<code>advanced_ml_strategies.py</code>	Advanced
S7	K-Means asset clustering + long-only momentum-alpha portfolio (top-2 of 5)	<code>advanced_ml_strategies.py</code>	Advanced

Each strategy exposes a common contract: `generate_signals(df)` returns a signal frame with consistent columns (`long_entry`, `short_entry`, `exit_long`, `exit_short`, ATR, Kumo edges) that both backtest engines and the live engine consume unchanged.

3.4 Execution & risk (`risk.py`)

- **Cost model:** taker/maker fee (0.1% default), half-spread (1 bp), slippage (2 bps) and square-root market impact $I = \xi \cdot \sigma \cdot \sqrt{(Q/V)}$ (Basilico et al.); shorts carry financing.
- **Position sizing:** fixed-fractional (risk-per-trade = 1% of equity) or half-Kelly.
- **Stops:** ATR or Kumo initial stop, optional trailing, conservative intrabar rules for gaps and both-levels-bars.
- **Circuit breakers:** max-drawdown and daily-loss limits halt the engine.

3.5 Backtesting engines

	Vectorised	Event-driven
Purpose	parameter sweeps	realistic evaluation
Sizing	full notional	risk-based
Stops	none	ATR/Kumo/trailing
Fill	signal-bar close (approx.)	next-bar open (no look-ahead)
Costs	proportional	fees + spread + slippage + impact
Breakers	none	max-DD / daily-loss

The event-driven engine mirrors the live engine's decision loop so that `backtest == live replay` as closely as possible (a dedicated parity test exists).

3.6 Real-time engine (`live_trader.py`)

Asyncio engine over ccxt.pro WebSocket / REST / replay feeds. Python paper execution walks the real order book (VWAP over consumed levels); CCXT enables testnet/live. Fills, trades, equity and events are journaled to SQLite.

3.7 Analytics (`analytics.py`)

Sharpe/Sortino/Calmar ratios, drawdown depth & **duration**, probabilistic Sharpe, VaR/CVaR, alpha/beta, trade statistics (win rate, profit factor, payoff ratio) and Matplotlib dashboards.

4. Validation Protocol

53 unit tests (`pytest`) enforce the project's honesty guarantees:

- **Causality:** RSI/divergence, momentum and regime labels never use future bars; `generate_signals` on a truncated prefix equals the first rows of the full frame.
- **Behaviour:** RSI divergence detects constructed patterns; K-Means regime labels are bounded and causal; cointegration analyzer recovers a known hedge ratio; ADF flags stationary series; Hurst separates trending from mean-reverting series; Bollinger bands keep 2σ geometry; calendar buckets use expanding means only; MomentumAlpha portfolio rebalances within top-N and rejects empty universes.
- **Config:** misconfigurations (no calendar features, empty universe) raise `ConfigError`.
- **Research/production parity:** replaying historical bars through the asynchronous live engine reproduces the event-driven backtest trade for trade (side, entry/exit timestamps, exit reason), so the configuration chosen in research is the configuration that trades.

Parameter-selection protocol. Every tunable choice reported in Section 5.5 is made on a chronological 70/30 in-sample/out-of-sample split, repeated on a second independent data window, and cross-checked on a second asset. A configuration is adopted only if it is positive in *both* halves of *both* windows. The single best in-sample cell is explicitly rejected: the in-sample optimiser's winner collapses out of sample (IS Sharpe $\sim 0.49 \rightarrow$ OOS ~ -2.7).

5. Results

Headline. Results are reported in two states, because the difference between them is the paper's main empirical point. *Original defaults* (1h bars, fixed 4xATR take-profit, no trailing stop) are in Section 5.1: every single-symbol strategy loses money. *Repaired defaults* (4h bars, no take-profit, Kijun-sen trailing stop), selected out-of-sample in Section 5.5, are in Section 5.6: over three years of BTC/USDT two of the five strategies turn positive (Ichimoku +19.3%, momentum +36.2%) and every strategy's drawdown shrinks. Neither state beats buy-and-hold over a tripling market (+197.3%), and the positive results remain thin, so Section 5.5 also reports their limits.

All results are event-driven, with costs (0.1% fee + 1 bp half-spread + 2 bps slippage = $\sim 0.1\%$ /side through the cost model) and \$10,000 initial capital.

5.1 Single-symbol strategies — BTC/USDT 1h, 180 days

Benchmark: buy-and-hold **+8.95%**, final **10,880.95**, annual vol 39.3%, Sharpe 0.64.

Strategy	Final	Return	CAGR	Sharpe	MaxDD	Time λ	Trades	Win%	PF
S4 Aroon/RSI divergence	9,052.62	-9.47%	-18.3%	-2.56	-10.96%	7.8%	63	33.3	0.56
S1 Ichimoku 10/30/60/30	9,637.03	-3.63%	-7.2%	-0.81	-5.48%	13.2%	28	32.1	0.74
S2 Momentum 20b	9,345.11	-6.55%	-12.8%	-1.07	-11.45%	18.3%	58	36.2	0.74

Strategy	Final	Return	CAGR	Sharpe	MaxDD	Time λ	Trades	Win%	PF
S3 Calendar anomalies	1,475.42	-85.25%	-97.9%	-21.1	-85.25%	29.4%	798	21.3	0.21
S6 Hurst + RSI	5,840.83	-41.59%	-66.4%	-14.3	-41.59%	5.1%	226	18.1	0.14
S5 Pairs (BTC×ETH, 60d win, z=2)	7,233.03	-27.7%	—	-0.77	-35.03%	—	121	31.4	0.71

Pairs was backtested over a longer 310-day window (buy-and-hold -6.9%, final 9,295.91), so its *Final equity* column is not directly comparable with the other rows.

Observations:

- **All long-only singles lost money** in a mildly rising sample because they were out of market most of the time (Time-in-market 5–29%) and their entries clustered in the noisy middle of the window. Costs dominate: the calendar strategy paid ~\$7,800 in fees+slippage (5,868 + 1,933) on \$10,000 capital across 798 trades — its Sharpe collapses to -21.
- **S1 Ichimoku** had the smallest loss and the best drawdown control (-5.5% vs -29.4% benchmark), consistent with its built-in cloud trailing. This row uses the *original* defaults (1h bars, fixed 4xATR take-profit, no trailing stop); Section 5.5 shows that those two choices, rather than the Ichimoku logic itself, caused most of the loss.
- **S6 Hurst** and **S3 Calendar** show the cost of over-trading in noise; both are pattern-laden but commercially unproven.

5.2 K-Means + momentum-alpha portfolio (S7)

Universe: BTC, ETH, SOL, ADA, XRP (1h, 120 days). K-Means (k=3) clusters assets by annualised volatility / momentum / log-volume; the engine ranks clusters by momentum and holds the top-2 assets equal-weight, rebalancing every 10 bars.

Metric	Portfolio	Buy & hold BTC/USDT
Final equity	8,083.26	9,574.27
Outperformance	-15.6%	—
Rebalances / trades	348	—

In the same window the altcoins polarised: ETH and SOL rallied (+11% each), while ADA fell ~22% and XRP ~7%. The equal-weight rule split the difference — it beat the two laggards it might have held but trailed the single BTC hold, so the experiment did **not** demonstrate alpha. Its value is the *plumbing*: quotas, rebalancing loop, and honest comparison against a real benchmark (the earlier +3.4% "beating buy-and-hold" figure was traced to a bug that compared the portfolio against ADA/USDT instead of BTC/USDT, and was corrected).

5.3 Live paper trading

The engine ran **live overnight** on BTC/USDT 1m (Sep 11 13:18 → Sep 12 02:37 UTC, ~13 hours, 800 equity samples): it opened 5 positions on closing bars — 4 were stopped out and 1 hit take-profit (net -\$162), matching the backtest's behaviour. Across the whole journal (paper + replay + live sessions) the engine recorded **57 round-trip trades / 116 fills**; the spread of results was wide (best single trade +\$222, worst -\$159) and the journal's total net P&L is -\$869. These are plumbing demonstrations — correct execution and honest bookkeeping — not evidence of a profitable edge. This validates the execution engine (fills, stops, financing, journaling) as live-sim ready.

5.4 Portfolio cluster details (real data, 120d)

Cluster	n_assets	mean_vol	mean_momentum	members
0	2	4.28	+0.018	ETH/USDT, SOL/USDT

Cluster	n_assets	mean_vol	mean_momentum	members
1	2	6.52	+0.0004	ADA/USDT, XRP/USDT
2	1	2.66	+0.009	BTC/USDT

5.5 Timeframe and exit-rule selection (out-of-sample sweep)

Sections 5.1-5.4 evaluate the strategy set at its original defaults. To separate *strategy logic* from *implementation choices*, `scripts/timeframe_sweep.py` backtests 384 configurations - timeframe (1h, 2h, 4h, 6h, 12h, 1d) x preset (standard, crypto, crypto_slow, dynamic) x TK-cross window (1, 3 bars) x exit rule (fixed take-profit with or without trailing; trailing-only with Kijun or Chandelier/ATR) x stop width (2, 4 ATR) - on three years of BTC/USDT (5 Oct 2023 - 9 Oct 2026, 26,399 hourly bars resampled upward; buy-and-hold +197.3% with a -53.5% drawdown). Each run is split 70/30 in-sample / out-of-sample, and the whole sweep is repeated on an independent two-year window.

Finding 1 - the fixed take-profit is the dominant leak. Trend systems earn from a thin right tail; a 4xATR target truncates exactly those trades while losers still run to the stop.

Exit rule	Median out-of-sample return
no take-profit + Kijun trailing stop	+2.9%
no take-profit + Chandelier (ATR) trailing	0.0%
take-profit + Kijun trailing	0.0%
take-profit, no trailing (original default)	-1.2%

Finding 2 - hourly bars lose to costs and whipsaw. With the exit rule above:

Timeframe	Median trades (3y)	Median OOS return	Median OOS Sharpe	Median OOS drawdown
1h	224	-3.4%	-0.24	-14.5%
2h	101	+8.8%	1.42	-5.3%
4h	52	+3.9%	0.77	-4.3%
6h	40	-1.9%	-0.63	-3.5%
12h	15	+3.8%	1.41	-1.5%
1d	9	+1.4%	0.95	-1.1%

2h scored highest, but 4h with the `crypto` preset was the only combination positive in all four segments (both halves of both windows) with a usable trade count; 12h and 1d trade 9-15 times in three years, too small a sample to trust.

Adopted configuration and effect. 4h, preset 10/30/60/30, 3-bar TK-cross window, no take-profit, Kijun-sen trailing stop, 2xATR stop:

	Original (1h, TP, no trail)	Adopted (4h, Kijun trail, no TP)
Total return (3y)	-33.9%	+19.3%
In-sample / out-of-sample	-17.8% / -19.6%	+9.5% / +9.0%
Max drawdown	-37.2%	-13.0%
Sharpe / Sortino	-	0.77 / 1.31
Profit factor	0.4-0.7	1.57
Trades / win rate	240 / 32.1%	67 / 26.9%
Time in market	-	13.3%
Annualised alpha vs buy-and-hold	-	+3.5%

Year by year (strategy vs buy-and-hold): 2023 (Oct-Dec) -3.0% vs +53.2%; 2024 +14.9% vs +121.1%; 2025 -5.0% vs -6.6%; 2026 (Jan-Oct) **+12.7% vs -6.4%**. The system is in the market 13.3% of the time, so it cannot track a tripling market; its contribution is drawdown control and positive performance when the trend turns.

Limits of this result, stated explicitly.

1. *Profit concentration.* Of 67 trades, 18 win and 49 lose; gross profit +5,336 against gross loss −3,401 on 10,000 of capital. The best single trade (17-26 Aug 2026, BTC 64.2k -> 78.4k) returns 1,937, while the whole three-year net profit is 1,934. Removing that trade leaves the period flat; removing the best three leaves it at −1,432. A long right tail is the intended behaviour of trend following, but with 67 trades the headline figure carries little statistical weight.
2. *Cross-asset decay.* The same configuration on three years of ETH/USDT 4h returns +4.6% against +52.4% buy-and-hold, with a −13.1% drawdown against −68.0%: the drawdown benefit transfers, the return does not.
3. *A tempting configuration was rejected.* A Chandelier (ATR) trailing stop scores far better on BTC (+34.8%, profit factor 1.90, best trade only 53% of profit) but loses on ETH (−3.4%, profit factor 0.90). Under the protocol of Section 4 it was not adopted, which is precisely the discipline the sweep exists to enforce.

5.6 The same repair applied to the whole strategy set

The exit rule and bar size of Section 5.5 are not Ichimoku-specific: every strategy in this framework shares `risk.py`. Re-running the full scorecard on the repaired defaults over three years of BTC/USDT 4h (python `scripts/compare_all.py --symbol BTC/USDT --timeframe 4h --days 1100`; buy-and-hold +197.3%, maximum drawdown −53.5%):

Strategy	Final	Return	CAGR	Sharpe	MaxDD	Trades	Win%	PF
S2 Momentum 20b	13,623	+36.2%	+10.8%	0.94	−20.6%	248	28.2	1.36
S1 Ichimoku 10/30/60/30	11,934	+19.3%	+6.0%	0.77	−13.0%	67	26.9	1.57
S4 Aroon/RSI divergence	9,031	−9.7%	−3.3%	−1.42	−9.9%	108	36.1	0.48
S6 Hurst + RSI	7,547	−24.5%	−8.9%	−2.10	−26.4%	307	33.2	0.54
S3 Calendar anomalies	1,916	−80.8%	−42.2%	−4.42	−82.1%	1,839	32.5	0.58

Compared with Section 5.1 (same strategies, original defaults, 180-day 1h sample), momentum moves from −6.6% to +36.2% and Ichimoku from −3.6% to +19.3%, while the two high-turnover strategies stay deeply negative: calendar anomalies trade 1,839 times in three years and pay their entire capital away in costs. Turnover, not signal sophistication, separates the two groups — and the ordering is the same conclusion the sweep reached for timeframes.

Two caveats carry over. The samples are different lengths (180 days versus three years), so the comparison is directional rather than like-for-like; and momentum's +36.2% comes with a −20.6% drawdown, well above Ichimoku's −13.0%, so on risk-adjusted terms the two are closer than the headline returns suggest (Sharpe 0.94 versus 0.77).

Tooling note. `scripts/compare_all.py` passed its argument list to `build_config()` instead of the parsed namespace, so `--symbol/--timeframe/--days` were silently discarded and every scorecard ran on the default window. The bug is fixed; the table above is the first scorecard whose window matches its caption.

6. Discussion

What the framework proves. The engineering contribution is real and measurable: causal indicators, two backtesting engines with matching semantics, a realistic cost/risk layer, and a live engine with verified paper execution. This is exactly what a *systematic* crypto pipeline must ship before any strategy can be trusted.

What the results say about retail strategies. The single-symbol results at the original defaults are a sober and useful negative result. In a mildly rising 180-day BTC sample, every long-only timing strategy lost money, and turnover relative

to costs predicts the outcome (calendar ■ hurst ■ momentum $\approx$ ichimoku $\approx$ divergence in cost discipline). The portfolio result is the encouraging counterpoint: risk-diversification across clustered assets and cross-sectional ranking traded better than timing a single asset.

How much of the loss was the strategy, and how much the implementation. Section 5.5 answers this directly, and the answer is uncomfortable for the usual presentation of indicator strategies: two implementation choices — bar size and exit rule — moved the same Ichimoku logic on the same asset from -33.9% to $+19.3\%$ over three years and cut the drawdown from -37.2% to -13.0% . Neither choice touches the signal. Published indicator results that do not state the bar size, the exit rule and the cost model are therefore close to uninterpretable. Equally, the repaired configuration does not establish an edge: its profit sits in one trade, its magnitude does not transfer to ETH, and over the full three years it remains far behind buy-and-hold. The defensible claim is narrower — the framework can tell a cost artefact from a signal, and it reports when the signal is thin.

Threats to validity. (i) two markets (BTC, ETH) and, for the selection study, two overlapping windows — still one exchange and one three-year sequence of regimes; (ii) the adopted configuration rests on 67 trades, one of which carries the entire profit, so its confidence interval is wide and no claim of significance is made; (iii) resampling 1h bars upward assumes a gap-free hourly history, which the data-quality report verifies for this sample but which would not hold through an exchange outage; (iv) single-seed synthetic tests only; (v) K-Means label non-determinism mitigated via `scipy kmeans2` seed + cluster-centre sorting; (vi) scikit-learn's KMeans was DLL-blocked by Windows Application Control on the dev machine, so the shipped code prefers sklearn and falls back to scipy with identical output.

7. Conclusion & Future Work

We presented a research-grade, honest, end-to-end framework for systematic crypto trading, implementing a full intermediate/advanced strategy set and demonstrating it on real data and in live paper trading. The contribution is validated infrastructure for cost-aware, causal evaluation — and a candid record that single-symbol retail timing strategies, once costs are included, do not simply work. The selection study of Section 5.5 sharpens that record: bar size and exit rule dominated the published loss, and repairing them turned -33.9% into $+19.3\%$ with a quarter of the drawdown, yet the repaired system still earns its profit from a single trade and does not reproduce that magnitude on a second asset. Infrastructure that separates these cases — and that refuses the better-looking but non-transferable configuration — is the result we claim.

Future work: (0) extend the selection sweep to walk-forward re-fitting across several assets and exchanges, until the headline figure rests on enough round trips to carry statistical weight; (1) ML return prediction (XGBoost/LSTM) feeding the portfolio ranker; (2) mean-variance/Hierarchical Risk Parity allocation in place of equal-weight; (3) multi-timeframe and multi-market (ETH, SOL, BTC-settled) validation; (4) testnet orders on Binance Spot Testnet with live execution journaling; (5) walk-forward / transaction-cost-adjusted parameter optimisation.

8. References

1. V. Vidyamurthy, *Pairs Trading: Quantitative Methods and Analysis*. Wiley, 2004.
2. E. Gatev, W. Goetzmann and K. Rouwenhorst, "Pairs Trading: Performance of a Relative-Value Arbitrage Rule," *Review of Financial Studies*, 2006.
3. N. Jegadeesh and S. Titman, "Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency," *Journal of Finance*, 1993.
4. E. J. Peters, *Fractal Market Analysis* (rescaled-range Hurst), Wiley, 1994.
5. B. Mandelbrot, "The Variation of Certain Speculative Prices," *Journal of Business*, 1963.
6. J. MacQueen, "Some Methods for Classification and Analysis of Multivariate Observations," *Proc. 5th Berkeley Symp.*, 1967 (K-Means).
7. S. A. Dickey and W. A. Fuller, "Likelihood Ratio Statistics for Autoregressive Time Series with a Unit Root," *JASA*, 1979.
8. A. Damodaran, *Investment Philosophies* (momentum & calendar effects), Wiley.
9. Basilico/E. for square-root market impact; Binance API documentation, and CCXT library documentation.

Appendix — Reproduction

```
git clone https://github.com/sakshamverma2030/Algorithmic-Trading-Framework-in-Crypto-Python.git
pip install -r requirements.txt
python main.py fetch --symbol BTC/USDT --timeframe 1h --days 180
python scripts/compare_all.py --symbol BTC/USDT --timeframe 1h --days 180
python main.py portfolio --source exchange --timeframe 1h --days 120

# Section 5.5 - selection sweep and the adopted configuration
python main.py fetch --symbol BTC/USDT --timeframe 1h --days 1100
python scripts/timeframe_sweep.py          # 384 runs -> reports/timeframe_sweep_BTCUSDT.csv
python main.py backtest --days 1100       # adopted defaults: 4h, Kijun trail, no take-profit
python main.py backtest --symbol ETH/USDT --days 1100  # cross-asset check

python -m pytest          # 53 tests (the ADF test needs statsmodels installed)
```